Prologue

The Impossible Done

John Bell first encountered the mathematics of quantum physics as a university student in Belfast, and he was not happy with what he found. To Bell, quantum physics was a vague mess. “I hesitated to think it was wrong,” said Bell, “but I *knew* it was *rotten*.”

The godfather of quantum physics, Niels Bohr, talked about a division between the world of big objects, where classical Newtonian physics ruled, and small objects, where quantum physics reigned. But Bohr was maddeningly unclear about the location of the boundary between the worlds. And Werner Heisenberg, the first person to discover the full mathematical form of quantum physics, was no better. Bohr and Heisenberg’s approach to quantum physics—known as the “Copenhagen interpretation,” named after the home of Bohr’s famous institute—was pervaded by the same vagueness that Bell had found in his quantum physics courses.

Shortly before Bell graduated from university in 1949, he stumbled upon a book by Max Born, another architect of quantum physics. Born’s book, *Natural Philosophy of Cause and Chance*, made quite an impression on Bell—especially the discussion of a proof by the great mathematician and physicist John von Neumann. According to Born, von Neumann had proven that the Copenhagen interpretation was the only possible way of understanding quantum physics. So either the Copenhagen interpretation was correct or quantum physics was wrong. And, given the wild success of quantum physics, it seemed that Copenhagen and its vagueness were here to